Global Inequality in Diabetes and Hypertension Care: A Visual Analytics Study

Group 5

CS661 Big Data Visual Analytics

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Contents

1 Project Repository and Live Deployment

The dashboard is deployed and publicly accessible — it can be used directly in a browser, with no installation:

- **Live application:** https://global-health-dashboard-1.onrender.com/
- **Source repository:** https://github.com/VarshithSS/global_health_dashboard

The repository holds the complete codebase, the prepared datasets, and this documentation. The dashboard is a single-page Dash application; to run it locally instead:

- `pip install -r requirements.txt`
- Run `python app.py`, then open `http://127.0.0.1:8050/` (the port can be overridden with the `PORT` environment variable)

The repository is organised so that the analytical layer and the presentation layer are cleanly separated: `data/` holds the eight prepared CSV tables, `visualizations/` holds six pure chart-building functions (each a `DataFrame → Figure` transformation with no knowledge of the application), and `app.py` holds the shared state, theming, layout, and callback logic that binds them into one linked dashboard.

2 Introduction

Diabetes and hypertension are two of the most widespread chronic conditions in the world, and managing them well depends less on a single dramatic intervention than on whether a health system can do two things reliably: reach people with treatment, and then keep their condition under control over the long term. These are not the same achievement. A country can put a large share of its hypertensive population on medication and still fail to bring most of them to a controlled blood pressure — and the difference between those two numbers is one of the most policy-relevant quantities in chronic disease care.

The World Health Organization tracks treatment coverage and effective control coverage for these conditions across nearly every country in the world, broken down by sex, and going back to 1990. This makes it one of the most reliable and analytically rich slices of global health data available for visual analytics. This project builds an interactive, web-based visual analytics system on top of that data to explore a single question from six complementary angles: *where in the world are people being treated for diabetes and hypertension but not having it effectively controlled, and how has that gap evolved and differed by sex, region, and income level?*

A deliberate decision shaped the entire project. The full WHO Health Inequality Data Repository is large but sparse: many of its indicators cover only a handful of countries per year, are missing national-level values entirely, or are disaggregated along dimensions with very limited coverage. Rather than build an impressive-looking system riddled with empty views, we audited every indicator for completeness and scoped the project to the six that are verified complete. The result is a smaller but fully populated analytical surface — every view in the dashboard is backed by real data for every country and every year in range, and no chart in this report is drawn on top of a gap.

3 Objective

The primary objective is to make global inequality in chronic disease care legible — to take three decades of WHO estimates and turn them into an interface where a public health student, researcher, or policy analyst can see not just *that* inequality exists but *where* it lives and *which mechanism* produces it. Specifically, the project aims to:

- Map treatment and control coverage for diabetes and hypertension across 195 countries and three decades, and expose the size of the gap between the best- and worst-served populations.
- Quantify and visualise the male–female coverage gap, and determine whether it is closing or widening.
- Track how coverage has evolved from 1990 onward for countries, WHO regions, and World Bank income groups, and test whether growth has been equitably shared.
- Expose the *treatment-to-control cascade* — the "leaky pipeline" in which patients are treated but not effectively controlled — which is only measurable because both indicators exist for the same country-year.
- Separate the effect of a population's age structure from the genuine performance of its care system, by comparing crude against age-standardized rates.
- Deliver all of the above as one *linked* dashboard, in which a selection made in any view propagates to every other view, rather than as a set of independent charts.

4 Dataset and Preprocessing

4.1 Source and the Scoping Decision

The data source is the WHO Health Inequality Data Repository (Health Care System and Access module), published by the WHO Global Health Observatory. The full repository spans 197 countries, the years 1980–2023, and 33 indicators, at roughly 100,000 rows.

We audited every one of those 33 indicators for completeness before writing a line of visualization code. Most turned out to be unusable for a country-comparative dashboard: financial-hardship and health-workforce indicators cover only a handful of countries per year; several have no national-level values at all; and others are only broken down by dimensions (wealth quintile, age band) whose country coverage is very thin. Building a world map on any of them would have produced a chart that was mostly empty.

We therefore scoped the project to the six indicators below — all from the Non-Communicable Disease (NCD) family — which we verified to have *zero* missing estimates, near-complete country coverage, and continuous multi-decade year coverage. Together they account for 72,420 country–year–sex estimates, the dense and reliable core of the repository.

Table 1: The six in-scope indicators, verified complete. Counts are from the prepared dataset itself, not from the repository documentation.

Indicator	Code	Countries	Years	Missing
Diabetes treatment coverage (age-standardized)	`diab_tx_std`	195	1990–2022	0
Diabetes treatment coverage (crude)	`diab_tx_crude`	195	1990–2022	0
Hypertension treatment coverage (age-standardized)	`htn_tx_std`	195	1990–2019	0
Hypertension treatment coverage (crude)	`htn_tx_crude`	194	1990–2019	0
Hypertension effective control (age-standardized)	`htn_ctrl_std`	195	1990–2019	0
Hypertension effective control (crude)	`htn_ctrl_crude`	194	1990–2019	0

Every estimate carries a Male/Female sex breakdown, a WHO region (six regions: African, Americas, Eastern Mediterranean, European, South-East Asia, Western Pacific), and a World Bank income group (Low, Lower-middle, Upper-middle, High-income; five small territories are unclassified). The two indicator families end in different years — diabetes runs to 2022, hypertension to 2019 — which is a small asymmetry with real consequences for the interface, discussed in Section 6.3.

4.2 Derived Metrics

Three derived quantities do the analytical work in this project. Each was computed during preprocessing and stored alongside the raw values, so that the visualization layer never has to recompute them:

- **Absolute Sex Equity Gap (ΔSex)** = $\text{Rate}_{\text{Male}} - \text{Rate}_{\text{Female}}$. A negative value means coverage is higher for women.
- **Systemic Care Pipeline Leakage (ΔLeak)** = $\text{Treatment}_{\text{crude}} - \text{Control}_{\text{crude}}$ (hypertension). The share of the population treated but not effectively controlled — the clinical drop-off.
- **Demographic Variance Offset (ΔAge)** = $\text{Rate}_{\text{crude}} - \text{Rate}_{\text{age-standardized}}$. How much a country's raw figure is inflated or deflated purely by the age structure of its population.

ΔLeak in particular is only computable because the repository publishes treatment *and* effective control for the same country, year, and sex. That coincidence is what makes the cascade view (Section 7.4) possible, and it is the single most valuable property of this dataset.

4.3 Prepared Analytical Tables

Rather than have every view filter one enormous table at request time, the preprocessing stage materialises eight task-shaped CSV tables. Each is pre-shaped for the view that consumes it — long format where a view needs to filter, wide format where it needs to pair columns, pre-aggregated where it needs group statistics.

Table 2: The eight prepared tables and the views they serve.

Table	Rows	Shape / contents	Used by
`map_data.csv`	72,420	Long: one row per country×year×sex×indicator	Task 1
`sex_gap_data.csv`	36,210	Male/Female pivoted side by side, plus ΔSex	Task 2
`trend_country.csv`	12,870	Wide: one column per indicator code	Tasks 3, 6
`trend_region.csv`	1,116	Region × year × indicator means	Task 3
`trend_income.csv`	930	Income group × year × indicator means	Task 3
`cascade_data.csv`	11,640	Hypertension treatment + control + ΔLeak	Task 4
`region_income_summary.csv`	4,278	Region × income cells: mean, median, std, count	Task 5
`age_vs_crude.csv`	36,150	Crude/standardized pairs, plus ΔAge	Task 6

Preprocessing also standardised country naming to ISO-3 codes (so the choropleth never silently drops a country), verified that every indicator is on the same 0–100 percentage scale, and confirmed the zero-missing-value property that motivated the scoping decision in the first place.

5 Tools and Technologies Used

The system is built entirely in Python, with the visualization and application layers both served from one process:

- **Dash 4.4 (Flask)** — the web application framework. Provides the reactive callback graph that keeps the six views synchronised with the shared filter state, and serves the single-page interface.
- **Plotly 6.9** — all six visualizations. Chosen over D3.js (which the proposal had listed as an option) because every chart type the project needed — choropleth, dumbbell, multi-line time series, horizontal cascade, grouped bars with error bars, paired slope chart — was expressible directly in Plotly's graph objects, and because Plotly figures arrive in the browser with hover, zoom, pan, and click events already wired to Dash. Writing custom D3 components would have added a JavaScript build step and a second event system for no analytical gain.
- **pandas 3.0 and NumPy 2.5** — data aggregation, the derived-metric computations, and all in-callback filtering.
- **Kaleido** — static export of the Plotly figures reproduced in this report, so that the figures shown here are rendered by the same code paths the live dashboard uses.
- **Git / GitHub** — version control across the team.

Two technologies named in the proposal were deliberately dropped. **D3.js** was not needed, for the reasons above. **SQLite** was listed as an optional storage layer, but at 72,420 rows the entire working set fits comfortably in memory as pandas DataFrames; introducing a database would have added a query layer and a serialisation boundary without measurably improving response time. Instead, repeat requests are served from an in-process cache (Section 6.6). Both decisions are revisited in Section 11.

6 System Design and Interaction Model

The proposal committed to "a single-page, web-based interactive dashboard (not a long scrolling page)... all connected through shared filters... Selections made in one view will update the other linked views." That commitment, rather than the individual charts, is what drove the architecture. This section describes what was built to honour it.

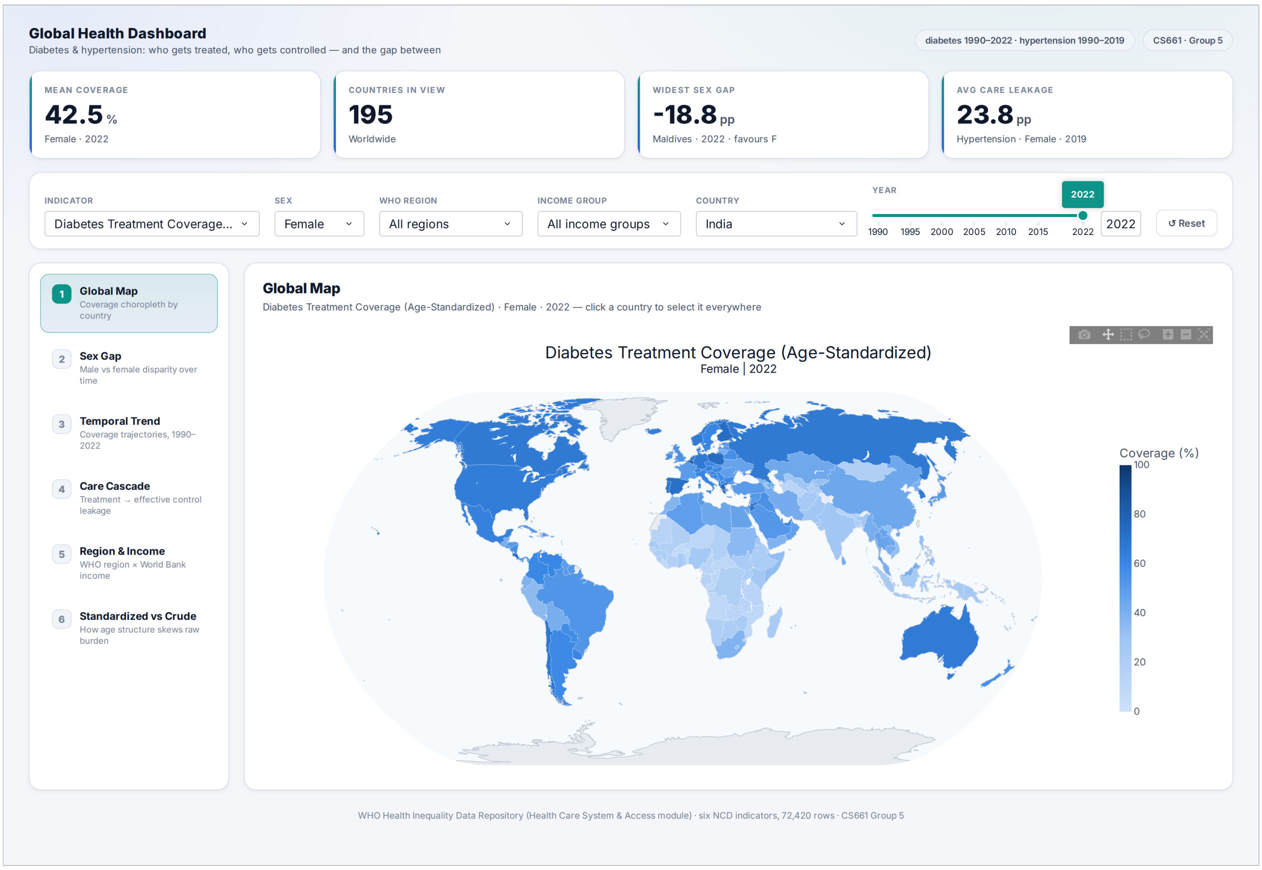

Figure 1: The dashboard interface. A brand header and live KPI strip sit above a sticky command bar carrying the six shared filters; a left navigation rail switches between the six analytical views without a page reload. Shown here in the default state: Task 1, diabetes treatment coverage (age-standardized), female, 2022.

6.1 Architecture

The code is split along a strict boundary. The `visualizations/` package contains six pure functions — `create_global_map()`, `create_sex_gap_chart()`, `create_temporal_trend_chart()`, `create_treatment_control_cascade()`, `create_regional_income_comparison()`, and `create_age_standardized_crude_chart()`. Each takes a DataFrame and a handful of scalar parameters and returns a Plotly figure. None of them import Dash, know that a dashboard exists, or read global state. Each validates its own inputs and raises on an impossible selection rather than silently returning an empty chart.

Above them, `app.py` supplies everything the charts deliberately do not know about: the shared filter state, the cross-filter logic, the light theme, the caching layer, and the layout. This separation is what allowed the six tasks to be built in parallel — each task's chart function could be written and tested against a CSV without the application existing, and the integration layer could restyle and re-link the entire dashboard without touching a single chart function.

6.2 Shared Filter State and Cross-Filtering

All six views read from exactly one source of truth: a `dcc.Store` holding six fields — indicator, sex, WHO region, income group, country, and year. A single reducer callback owns that store. It takes as inputs both the six controls in the command bar *and* the click events from five of the six charts, merges them, and writes the resulting state back out. Because every view's callback listens to the store rather than to the individual controls, adding a new cross-filter interaction never requires touching the views.

The cross-filter behaviours this produces are:

- **Click a country on the map** → sets the global country, which immediately re-draws the sex-gap chart, the care cascade, and the crude-vs-standardized comparison for that country.
- **Click a bar in the region×income view** → sets both the WHO region and the income group, narrowing the country pool everywhere.
- **Click a point on a trend line** → sets the year (and, at country level, the country), so a moment of interest in a time series becomes the selected year across the whole dashboard.
- **Click a year in the sex-gap chart** → sets the year.

Region and income act as a *scope*: they narrow the pool of countries the rest of the dashboard may consider. When the scope changes, the country dropdown is re-populated, and if the currently selected country has fallen outside the new scope it is automatically replaced with a sensible in-scope one rather than leaving the dashboard in an impossible state.

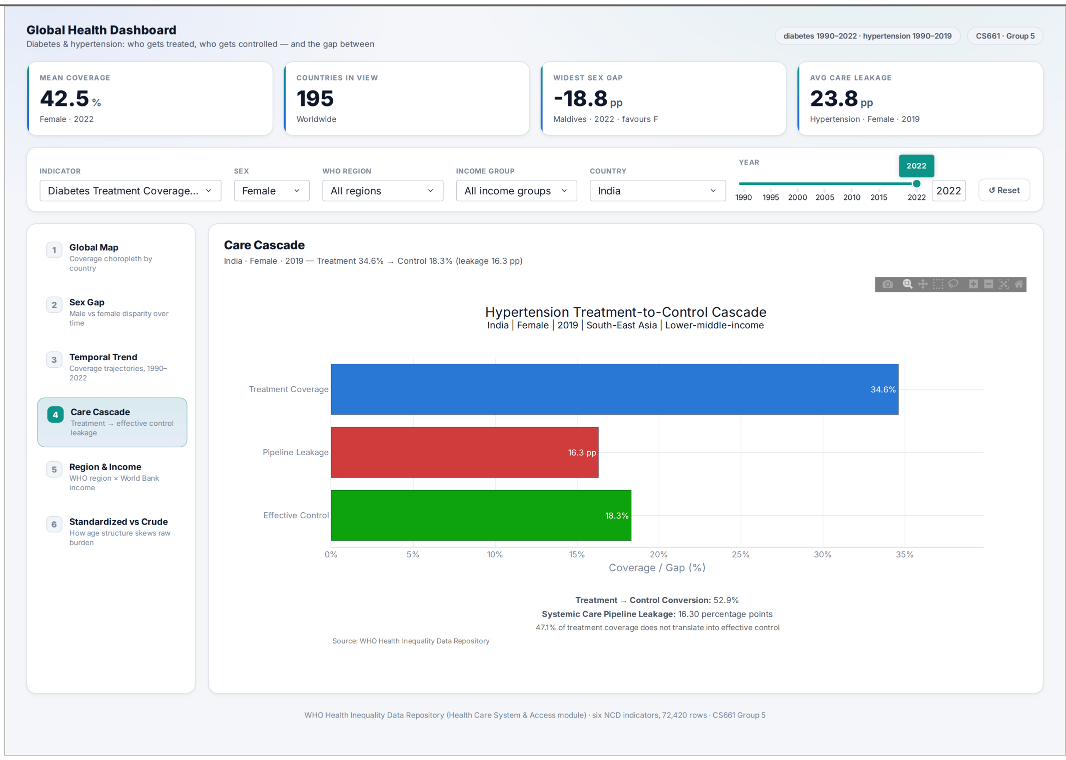

Figure 2: The same filter bar, a different view. The country here (India) was not typed in — it was inherited from the shared state. Every view reads the same six-field store, so a country selected on the map in Figure 1 is already selected here.

6.3 Nearest-Year Fallback

Because the diabetes indicators run to 2022 and the hypertension indicators stop at 2019, a naive implementation has an ugly failure mode: a user viewing diabetes data in 2022 who switches the indicator to hypertension would get a blank chart, for a reason the interface never explains. Rather than clamp the year slider (which would lose the user's position) or blank the view, the dashboard snaps to the *nearest available year* for the selected indicator and states plainly in the view's subtitle that it has done so: "no 2022 data; showing nearest year." The same helper backs the KPI strip, so the headline numbers never disagree with the chart below them. This is a small piece of logic, but it is the difference between an interface that feels broken and one that feels considered.

6.4 The KPI Insight Strip

Four headline numbers sit above the views and recompute on every filter change: mean coverage across the countries currently in scope, the number of countries in scope, the widest sex gap in scope (with the country and direction that produced it), and the mean treatment-to-control leakage. Their purpose is to give the current selection a summary before the user has read the chart — and, because they are computed over the *scope* rather than the single selected country, they provide the context against which the selected country should be judged.

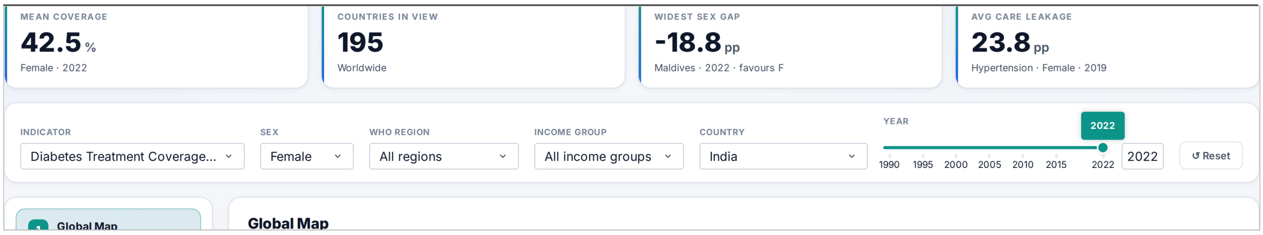

Figure 3: The KPI strip and the sticky command bar. Every number shown is recomputed from the current scope; here, worldwide, female, 2022.

6.5 Visual Design and Accessibility

The interface uses one cohesive light theme, applied to the charts *after* they are built, so the chart functions stay presentation-neutral. Every palette below was validated against the surface the charts actually render on (white), rather than chosen by eye. Three choices are worth recording:

- **Colour-blind-safe categorical palette.** The eight-colour series palette was chosen and ordered so that adjacent series remain distinguishable under common colour-vision deficiencies — the worst adjacent pair separates by ΔE 24.2 under protanopia, comfortably clear of the ≥ 12 threshold. Every categorical chart is additionally paired with a legend, so identity is never carried by colour alone.
  The trade-off we accepted: three of those eight hues (aqua, yellow, magenta) fall below the 3:1 contrast floor against white. This is intrinsic rather than an oversight — yellow has to stay *light* to separate from green for red-blind viewers, and darkening it to an ochre collapses that pair to ΔE 2.1, i.e. indistinguishable. We kept the colour-blind-safe hues and bought back the contrast with a second channel instead: every bar and marker carries a dark hairline edge, so a pale fill still reads against the white surface.
- **Semantic colour in the cascade.** Task 4 is the one view where colour encodes *state* rather than identity: treatment is a neutral blue, leakage is red (a loss), and effective control is green (the desired outcome). These are reserved status colours, never reused for a data series, and each bar is labelled — so the colour reinforces the meaning rather than carrying it. Using the categorical palette here would have thrown away that meaning.
- **A sequential ramp for the map.** The choropleth uses a single-hue pale-to-deep blue ramp on a fixed 0–100 domain, so that colours are comparable across years and indicators — animating the year slider shows real change, not a rescaled legend. Countries with no data are a neutral grey, deliberately distinct from the palest step of the ramp, so "no data" can never be misread as "near zero".

6.6 Performance

All eight tables are loaded once at start-up and held in memory. Each of the six figure builders is wrapped in an LRU cache keyed by the exact filter tuple that produced it, so dragging the year slider back and forth, or returning to a view already visited, is served from cache rather than re-filtering the DataFrame and re-building the figure. At this data size the result is an interface that responds to a filter change without a perceptible pause.

7 Visualization Tasks and Insights

The six views below are the analytical core of the project. Each is presented with the same structure: what the task does, what information it encodes, what we actually observed in the data, what can be inferred from it, and what it implies in the real world. Every number quoted in these sections was computed from the prepared tables described in Section 4.

7.1 Global Coverage Map

7.1.1 Task Overview

An interactive world choropleth showing the value of the selected indicator for every country, for a chosen year and sex. The year slider spans the full range of the selected indicator, so dragging it animates three decades of change. Clicking any country promotes it to the global country selection, making the map the primary entry point into the rest of the dashboard: the user starts with a global overview, spots an outlier, clicks it, and every other view is now about that country.

7.1.2 Information Represented

- **Colour:** coverage (%) on a fixed 0–100 sequential scale, so colour is comparable across years and indicators.
- **Geography:** 195 countries, keyed by ISO-3 code.
- **Hover:** exact coverage, WHO region, and World Bank income group.
- **Filters:** indicator (all six), sex, year; plus the region and income scope, which grey the map down to the countries under consideration.

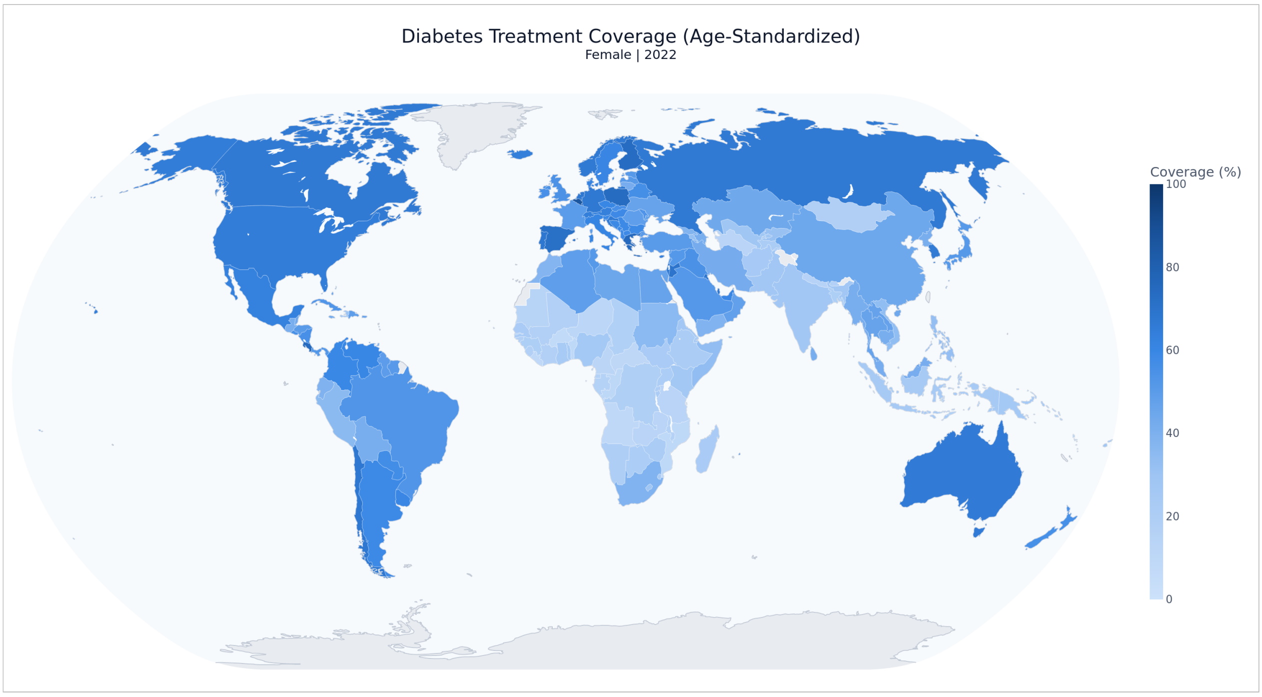

Figure 4: Diabetes treatment coverage (age-standardized), female, 2022. Deeper blue is higher coverage. The north–south gradient is immediate: Europe and the Americas are saturated, sub-Saharan Africa is uniformly pale.

7.1.3 Observed Trends

- For diabetes treatment (age-standardized, female, 2022) the global mean is **42.5%**, but the spread is enormous: from **8.5% in Burkina Faso** to **86.0% in Belgium** — a **10.2×** ratio between the worst- and best-served countries.
- The eight lowest-coverage countries are all in the WHO African region (Burkina Faso, Angola, Liberia, Benin, Mozambique, Central African Republic, Congo, Niger — all at or below 12%). The top of the table is European, with Costa Rica and Jordan the notable non-European entrants.
- Regional means separate cleanly: European 56.9%, Americas 51.5%, Eastern Mediterranean 47.4%, Western Pacific 39.1%, South-East Asia 33.6%, African **21.5%**.
- The income gradient is even cleaner than the geographic one: high-income 59.8%, upper-middle 44.5%, lower-middle 29.0%, low-income 22.7%.
- Switching the indicator to **hypertension effective control** (2019) collapses the whole map downward: the global mean falls to just **23.0%** for women, ranging from 5.1% (Indonesia) to 57.3% (Republic of Korea). For men it is lower still, at 16.4%.

7.1.4 Inference and Insights

- Chronic-disease care is not merely unequal — it is *stratified*. The ordering of regions and of income groups is stable across indicators, which means we are not looking at noise or at a handful of outliers but at a structural gradient.
- The single most important observation is what happens when the indicator is switched from treatment to *effective control*: the entire world map drains of colour. A global treatment mean in the mid-40s becomes a global control mean in the low-20s. Being treated and being controlled are demonstrably different achievements, which is precisely the gap Task 4 exists to measure.
- Even the best-performing country in the world controls only 57% of its hypertensive women. There is no country where this problem is solved.

7.1.5 Real-World Implications

- **Global health funders** can use the map to target the countries where absolute coverage is lowest — and the concentration of the bottom of the distribution in one WHO region makes that targeting unusually clear-cut.
- **National ministries** can locate themselves against regional and income peers rather than against a global average that mixes very different systems.
- **The control-coverage map is the more actionable of the two.** A country with high treatment and low control does not need a wider net — it needs better follow-up, adherence support, and titration. That is a different budget line, and the map makes the distinction visible at a glance.

7.2 Sex Gap Analysis

7.2.1 Task Overview

A dumbbell chart comparing male and female coverage for one country across every year in the indicator's range. Each horizontal connector spans the gap for a single year; the length of the connector *is* the inequity.

Plotting all years at once — rather than one year at a time — means the chart answers not only "how big is the gap?" but "is it closing?"

7.2.2 Information Represented

- **X-axis:** coverage (%), on a dynamically zoomed range so that a 2-point gap is still visible rather than being crushed against a 0–100 axis.
- **Y-axis:** year, one row per year.
- **Circle marker:** female coverage. **Diamond marker:** male coverage.
- **Connector length:** ΔSex = Male − Female.
- **Footer annotations:** the mean gap across all years and the largest single-year gap, each labelled with the direction it favours.

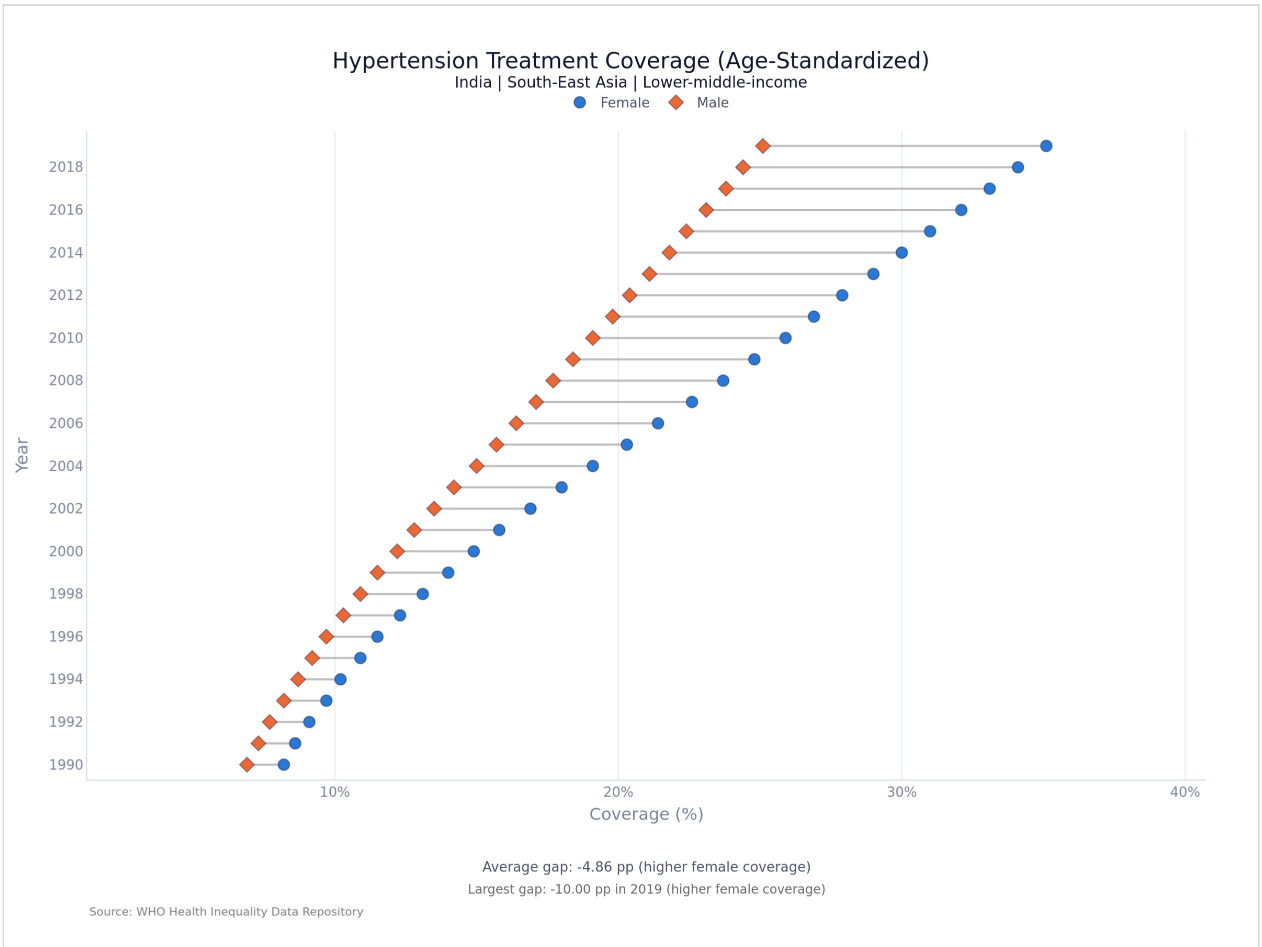

Figure 5: Hypertension treatment coverage (age-standardized), India, 1990–2019. Female coverage (circles) sits to the right of male coverage (diamonds) in every single year, and the gap widens over time.

7.2.3 Observed Trends

- The male–female gap in **hypertension treatment** is large and almost universal. In 2019 the mean ΔSex was **−11.4 pp**, and coverage was higher for women in **190 of 195 countries (97%)**. Only Canada (+5.2 pp), Norway, Kiribati, and Austria showed any male advantage at all.

- The same pattern holds for **hypertension effective control** (mean −6.6 pp; women ahead in 186 of 195 countries) and, more weakly, for **diabetes treatment** (mean −2.6 pp; women ahead in 131 of 195, or 67%).
- The extremes are dramatic: Saint Lucia (−28.0 pp), Jamaica (−26.4), Ecuador (−25.8) and Peru (−24.8) for hypertension treatment; the Maldives (−18.8 pp) for diabetes.
- **The gap is not closing — it is widening.** For hypertension effective control, the mean gap grew from −2.6 pp in 1990 to **−6.6 pp** in 2019. For hypertension treatment it went from −9.9 to −11.4 pp. Thirty years of progress have not narrowed the sex gap in either direction; they have stretched it.
- Regionally, the Americas show by far the widest female advantage (−17.6 pp for hypertension treatment). The African region is the only one where diabetes treatment slightly favours men (+1.2 pp).
- **India is a counter-example.** Its diabetes treatment gap favours men (mean +2.6 pp), but it is one of the few gaps in the dataset that is genuinely narrowing: from +3.7 pp in 1990 to +1.4 pp in 2022.

7.2.4 Inference and Insights

- A sex gap this consistent across 97% of countries cannot be explained by any individual country's culture or health policy. It points to a mechanism that operates almost everywhere — most plausibly, differential *contact* with the health system. Women in most countries interact with primary and reproductive health services far more often than men do, and each contact is an opportunity for an incidental blood-pressure measurement and a diagnosis. Men are not being denied treatment so much as never being diagnosed.
- This reframes the finding. The inequity is not that women are over-treated; it is that **men are systematically under-diagnosed**, and the coverage gap is the visible shadow of an invisible diagnostic gap.
- The widening trend supports this reading. As health systems have expanded since 1990, they have expanded along existing channels — and those channels reach women. Growth has amplified the asymmetry rather than correcting it.
- That the gap is much smaller for diabetes (−2.6 pp) than for hypertension (−11.4 pp) is consistent with this: diabetes is more likely to become symptomatic and force a clinical encounter, whereas hypertension is asymptomatic and therefore relies entirely on opportunistic screening.

7.2.5 Real-World Implications

- **Screening programmes should be designed to reach men where they already are** — workplaces, pharmacies, sporting venues — rather than waiting for them to present at clinics they rarely visit.
- **Health ministries** should treat a large negative ΔSex as a diagnostic coverage warning, not as evidence of successful women's health programming.
- **Equity monitoring frameworks** that report only national averages will miss this entirely: a country can raise its headline coverage while leaving half its population's diagnosis rate untouched.

7.3 Temporal Trend Analysis

7.3.1 Task Overview

A multi-line time-series chart tracking coverage from 1990 onward. The view supports three levels of aggregation through a "Compare by" control — country, WHO region, or World Bank income group — so the same chart can answer "how has India done?" and "have poor countries caught up?" without changing views. Clicking a point on any line sets the global year, feeding that moment back into the rest of the dashboard.

7.3.2 Information Represented

- **X-axis:** year (1990–2022 for diabetes, 1990–2019 for hypertension).
- **Y-axis:** coverage (%), auto-ranged to the visible series.
- **One line per entity** (up to several countries, six WHO regions, or four income groups), with a unified hover that reads every series at a given year at once.
- At country level the sex filter applies; the regional and income tables are pre-aggregated across sex.

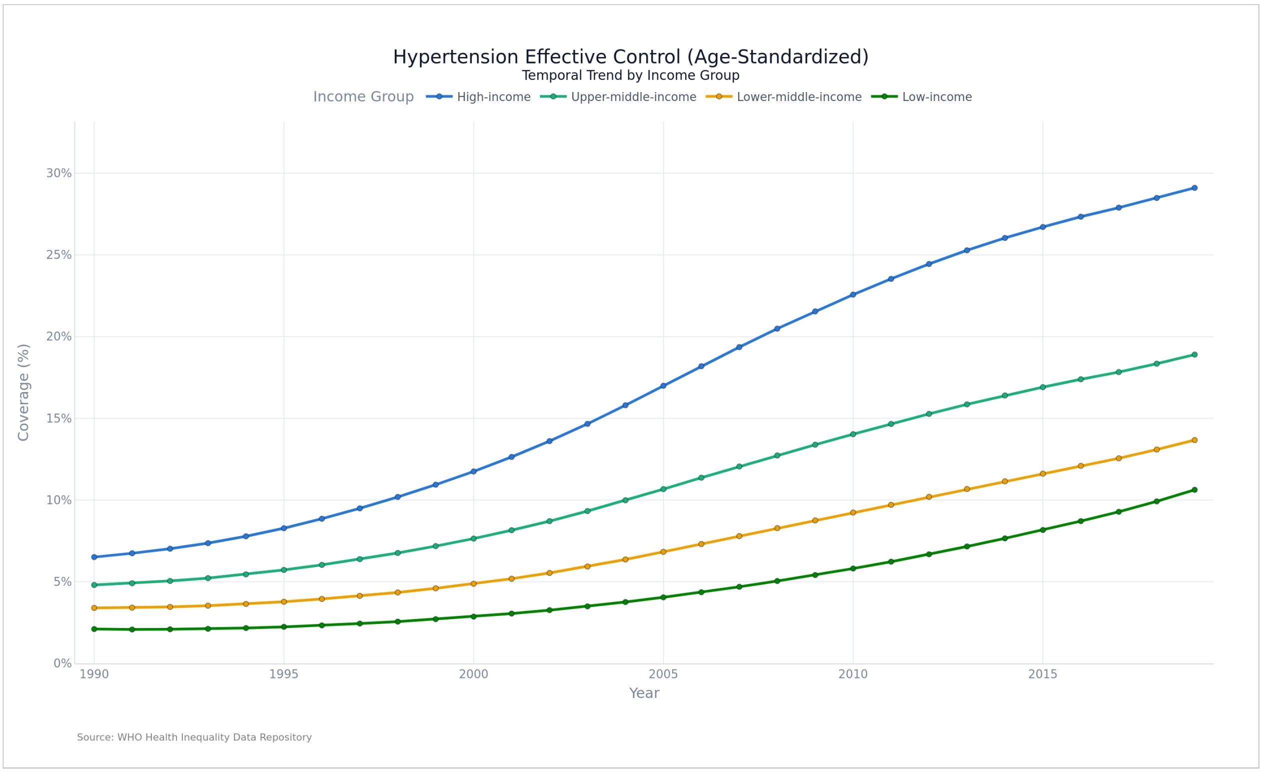

Figure 6: Hypertension effective control (age-standardized) by World Bank income group, 1990–2019. All four groups rise — and visibly fan apart. The vertical distance between the top and bottom lines in 2019 is more than four times what it was in 1990.

7.3.3 Observed Trends

- **Everything improved.** Every region and every income group ends higher than it started, for all three indicator families. Global diabetes treatment coverage rose from a mean of 30.2% in 1990 to 42.5% in 2022.

- **The relative gains were largest at the bottom.** Hypertension effective control in the African region grew by **437%** in relative terms (2.2% → 11.7%); in low-income countries it grew 405% (2.1% → 10.6%). A naive reading of percentage growth would conclude that the poorest countries are catching up fastest.
- **The absolute gaps widened in every single case.** This is the central finding of the view:

Table 3: Change in the high-income minus low-income coverage gap, 1990 to the latest available year. Every gap is larger today than it was in 1990.

Indicator	Gap in 1990	Gap in final year	Change
Diabetes treatment (age-std.), to 2022	20.4 pp	35.2 pp	+14.8 pp
Hypertension treatment (age-std.), to 2019	18.2 pp	28.0 pp	+9.7 pp
Hypertension effective control (age-std.), to 2019	4.4 pp	18.5 pp	+14.1 pp

- By region, European diabetes treatment coverage gained **+21.7 pp** (34.0% → 55.7%) while the African region gained only **+4.9 pp** (17.2% → 22.1%) over the same 32 years.
- Hypertension effective control in high-income countries gained +22.6 pp; low-income countries gained +8.5 pp.

7.3.4 Inference and Insights

> **Key finding.** Growth rates converged while levels diverged. The poorest countries improved *fastest in percentage terms* and yet fell *further behind in absolute terms* — because a 400% gain on a base of 2% is worth 8 percentage points, while a 350% gain on a base of 6.5% is worth 23. Percentage growth is a flattering and misleading measure of progress in global health; the honest one is the percentage-point gap, and by that measure inequality in chronic disease care has grown steadily worse since 1990.

- This is why the view exposes both the lines and the levels rather than plotting indexed growth. An indexed chart would have shown the African region as the global success story of the last three decades. The un-indexed chart shows it falling further behind every year.
- The effect is strongest for effective control — the hardest of the three indicators to deliver, and the one that most depends on a durable, funded, follow-up-capable health system rather than on a one-off intervention. Gaps widen fastest where sustained system capacity matters most.

7.3.5 Real-World Implications

- **Progress reporting that leads with percentage growth is actively misleading**, and this view makes the distortion visible. Global health targets should be set in percentage points, not percentage improvements.
- **Convergence will not happen on its own.** Thirty years of universal improvement have produced a wider gap, not a narrower one; closing it requires disproportionate, targeted investment rather than a rising tide.
- **For the SDG 3.4 agenda** (reducing premature NCD mortality by a third), the trajectory of the low-income line is the binding constraint, and it is the flattest line on the chart.

7.4 Treatment-to-Control Cascade

7.4.1 Task Overview

This is the view the whole project was scoped around. For a chosen country, year, and sex, it places hypertension *treatment* coverage directly against *effective control* coverage, and renders the difference between them as an explicit third bar: the leakage. It answers a question that neither indicator can answer alone — of the people this health system has reached, how many has it actually helped?

7.4.2 Information Represented

- **Three horizontal bars:** treatment coverage (%), pipeline leakage (pp), and effective control (%).
- **Semantic colour:** blue = treated (neutral), red = leaked (loss), green = controlled (the desired outcome).
- **Derived footer metrics:** the *treatment-to-control conversion rate* (control ÷ treatment) and the *leakage rate* (the share of treatment that fails to convert).
- The country is normally inherited from a click on the map rather than typed in.

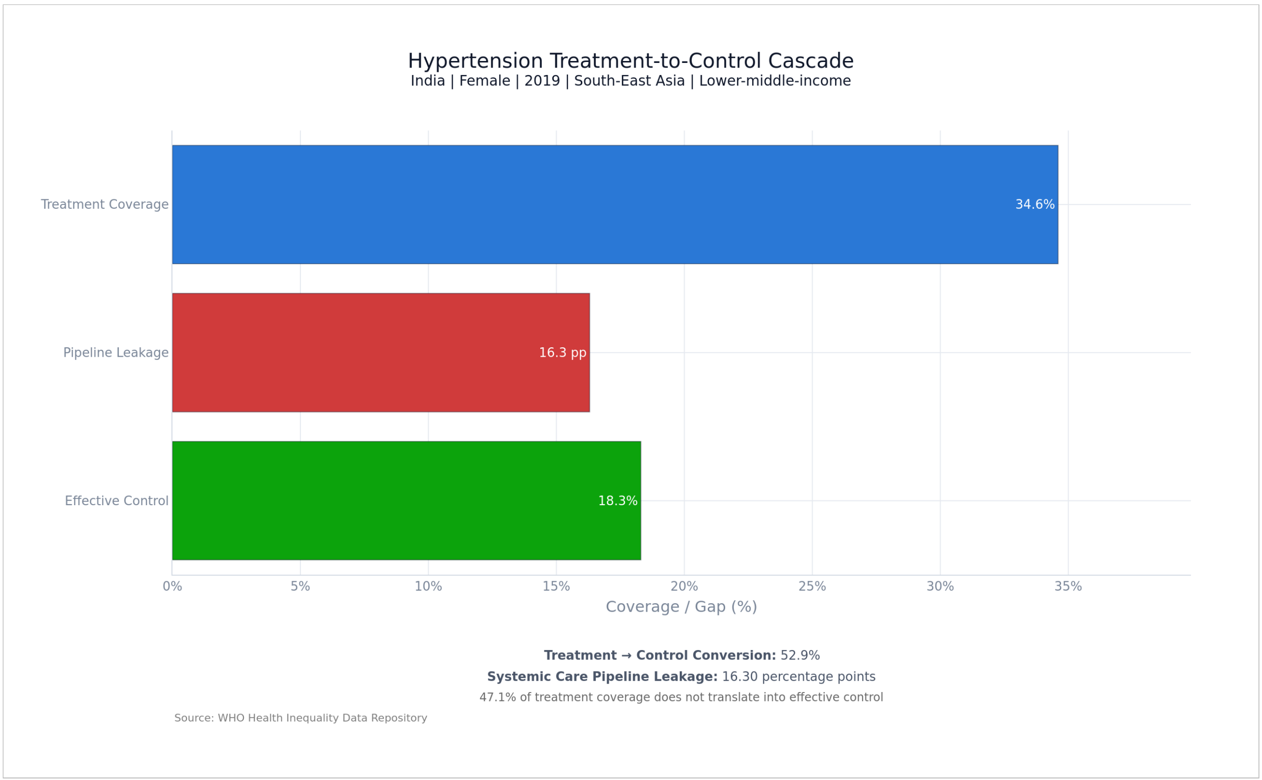

Figure 7: Hypertension treatment-to-control cascade, India, female, 2019. Of 34.6% treatment coverage, only 18.3 percentage points survive as effective control — a 16.3 pp leak, meaning 47.1% of treatment coverage does not translate into a controlled patient.

7.4.3 Observed Trends

- Globally, in 2019, mean female hypertension treatment coverage was **46.9%** and mean effective control was **23.1%** — a mean leakage of **23.8 pp**. The mean country-level conversion rate is **47.5%**: **fewer than half of all treated women reach control.** For men it is worse still, at 44.6%.

- Conversion varies enormously between health systems. Canada converts **79%** of treated women (and 84% of treated men) into controlled patients; the Republic of Korea 74%, Germany and Iceland 72%. At the other end, the Republic of Moldova converts **21%**, Belarus 23%, Kyrgyzstan 24%, and Indonesia 25%.
- The largest absolute leaks are concentrated in a striking geographic cluster: Belarus (46.5 pp), Ukraine (45.3), Latvia (41.5), Lithuania (41.3), and Croatia (40.9) — post-Soviet health systems that treat a great many people and control comparatively few of them.
- India (female, 2019) sits close to the global median: 34.6% treated, 18.3% controlled, a 16.3 pp leak and a 52.9% conversion rate.
- Over time, the pipeline has improved: in 1990 global mean treatment was 24.8% and control 6.0% (a conversion rate of roughly 24%); by 2019 treatment had nearly doubled to 46.9% and control had nearly quadrupled to 23.1%.

7.4.4 Inference and Insights

> **The leakage paradox.** High-income countries have the *largest absolute leak* (26.5 pp, versus 15.6 pp in low-income countries) and simultaneously the *best conversion rate* (54.7% versus 44.2%). These are not in conflict — they are arithmetic. Leakage measured in percentage points scales with how many people you treat in the first place, so a country that treats almost nobody cannot leak much. Ranking health systems by absolute leakage would therefore reward the countries doing the least. The conversion *rate* is the fair comparison, and the dashboard reports both precisely so that the naive metric cannot be read in isolation.

- By that fair measure, the worst performers are not the poorest countries but the **upper-middle-income** ones (43.7% conversion — the lowest of any income group, below even low-income countries at 44.2%). These are systems that have successfully scaled up *prescribing* without scaling up the follow-up, monitoring, and titration that convert a prescription into a controlled patient.
- The post-Soviet cluster is the clearest instance of this: high treatment reach inherited from a strong primary-care tradition, combined with weak chronic-disease follow-up, produces the worst leaks in the world.
- The finding generalises: **getting a patient onto medication is the easy half of the problem.** Globally, more than half of all the effort spent on treating hypertension does not currently result in a controlled patient.

7.4.5 Real-World Implications

- **For health ministries:** a large leak is a signal to invest in adherence support, dose titration protocols, and routine follow-up — not in wider prescribing. Those are cheaper interventions than expanding treatment reach, and this view identifies exactly which countries would benefit most from them.
- **For global health targets:** treatment-coverage targets can be met while health outcomes barely move. Control coverage is the outcome that matters, and it should be the number that is targeted.
- **For clinicians and system designers:** the countries with the best conversion rates (Canada, Republic of Korea, Germany, Iceland, Portugal) are the natural places to look for transferable follow-up models.

7.5 Regional and Income-Group Comparison

7.5.1 Task Overview

A grouped bar chart crossing the six WHO regions against the four World Bank income groups, for a chosen indicator and year. Its purpose is to disentangle two explanations that are usually confounded in global health commentary: is a country's coverage determined by *where it is*, or by *how rich it is*? Error bars carry the within-cell standard deviation, so the chart also shows how much internal disagreement each bar is hiding.

7.5.2 Information Represented

- **X-axis:** WHO region. **Bar groups:** World Bank income group, ordered low to high.
- **Bar height:** mean coverage for the countries in that region×income cell.
- **Error bars:** ±1 standard deviation within the cell.
- **Hover:** mean, median, standard deviation, and the number of countries in the cell — so a bar backed by two countries cannot be mistaken for one backed by twenty.
- Clicking a bar sets both the region and the income scope for the whole dashboard.

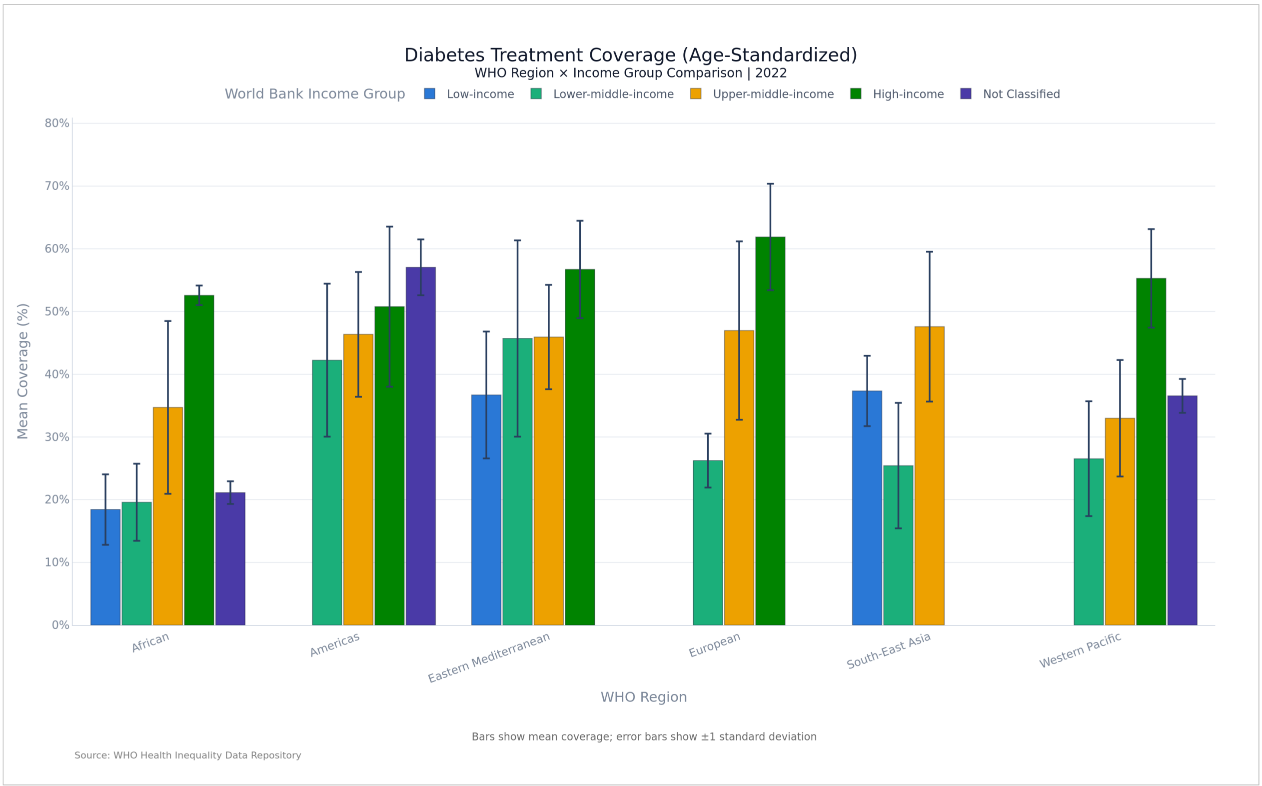

Figure 8: Diabetes treatment coverage (age-standardized), 2022, by WHO region and World Bank income group. Within every region, the bars climb with income — the income gradient survives geography.

7.5.3 Observed Trends

- **The income gradient holds inside every single region.** In no region does a poorer income group outperform a richer one for diabetes treatment.

- The gradient *within* the African region is larger than the gap *between* most regions: African low-income countries average **18.4%**, African high-income countries average **52.6%** — a spread of **34.2 pp** inside one region.
- Cross-region comparisons at matched income levels are the revealing ones. A **high-income African** country (52.6%) has roughly **double** the coverage of a **lower-middle-income European** country (26.2%). Geography loses; income wins.
- Overall, high-income countries average 55.5% against low-income countries' 30.8% — a 1.8× ratio.
- The error bars are large. Mean within-cell standard deviation is 8.4 pp, and the most heterogeneous cell (Eastern Mediterranean, lower-middle-income, 16 countries) has a standard deviation of 15.7 pp — wider than the difference between several regional means.

7.5.4 Inference and Insights

- **Income group is a stronger predictor of chronic-disease care than WHO region is.** The regional ordering seen on the map in Task 1 is, to a substantial degree, an income ordering wearing a geographic costume — the African region looks worst partly because it contains most of the world's low-income countries.
- But region is not merely a proxy for income, and the chart shows where it is not. African low-income countries (18.4%) still trail Eastern Mediterranean low-income countries (36.7%) by a wide margin at the same income level, which means something other than national income — health system structure, conflict, workforce density — is doing real work there.
- **The error bars are the honest part of this chart.** With a within-cell standard deviation of 8–16 percentage points, the mean of a region×income cell is a weak summary of the countries inside it. Any policy that treats "lower-middle-income Africa" as a homogeneous target is aggregating over countries that differ from each other by more than the group differs from its neighbours. This is the strongest argument in the dashboard for drilling down to the country level — which is what clicking a bar does.

7.5.5 Real-World Implications

- **Development finance** keyed to income classification is, on this evidence, better aligned with need than aid keyed to region.
- **Regional health bodies** should not benchmark member states against a regional mean; a country should be compared against its income peers, which this view makes possible in one click.
- **The heterogeneity within cells** is a warning to anyone building a one-size-fits-all regional intervention: the average country in a cell may not exist.

7.6 Age-Standardized vs. Crude Rate Comparison

7.6.1 Task Overview

Every indicator in this dataset exists in two forms: a *crude* rate, which reports what is actually happening in the country's real population, and an *age-standardized* rate, which re-weights that population to a common reference age structure so that countries can be compared fairly. This view puts the two side by side as a paired slope chart. A steep line means a country's raw numbers are being substantially distorted by its demography; a flat line means its crude figure can be taken at face value.

7.6.2 Information Represented

- **Two x-positions:** crude rate (circle) and age-standardized rate (diamond).
- **One coloured line per country,** connecting its two values — the slope is ΔAge = crude − standardized.
- **Footer:** the mean ΔAge across the selected countries and the single largest absolute offset.
- The view keeps its own "indicator family" control, because it pairs crude and standardized columns rather than treating them as separate indicators.

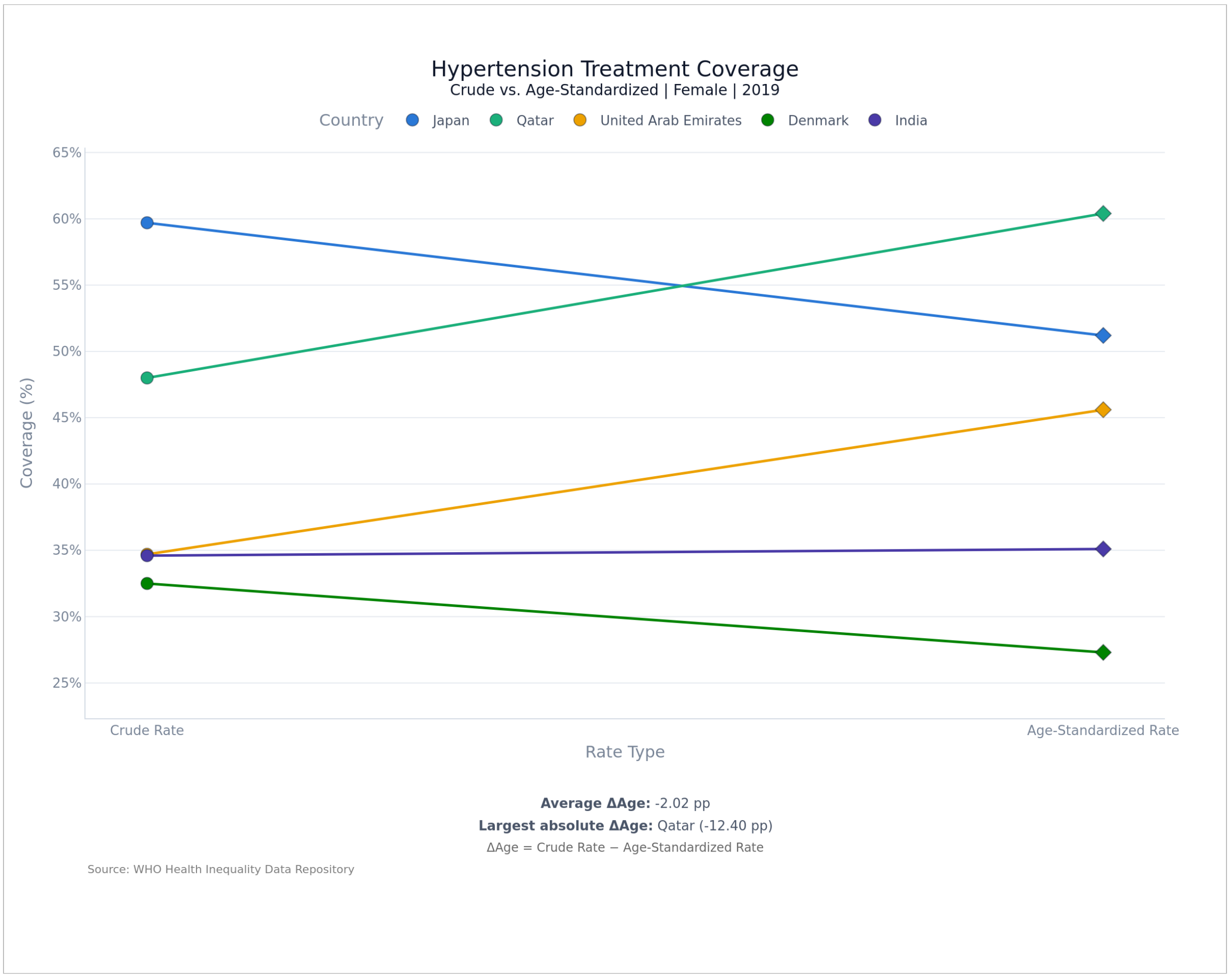

Figure 9: Hypertension treatment coverage, female, 2019. Japan and Qatar cross: by crude rates Japan (59.7%) leads Qatar (48.0%) comfortably, but once age structure is controlled for, Qatar (60.4%) overtakes Japan (51.2%). The two countries swap places purely because of demography.

7.6.3 Observed Trends

- Globally the two measures agree closely — the correlation between crude and age-standardized diabetes treatment coverage across 195 countries is **r = 0.994**. For most countries the choice of measure barely matters.
- But for demographic outliers it matters enormously. The **Gulf states** have the largest negative offsets — Qatar (−12.4 pp for hypertension treatment), the UAE (−10.9), Oman (−7.1), Kuwait (−6.3) —

because their populations are dominated by young migrant workers, which drags the crude rate far below the age-standardized one.

- The largest positive offsets belong to **aged populations**: Japan (+8.5 pp), Lithuania (+5.3), Latvia (+5.3), Bosnia and Herzegovina (+5.3), Denmark (+5.2), Sweden (+5.1). Their crude rates flatter them.
- **The rank consequences are severe.** Ranking all 195 countries by hypertension treatment coverage, **Qatar sits at #43 on age-standardized rates and #97 on crude rates — a 54-place swing.** Japan moves 28 places in the opposite direction (#87 standardized → #58 crude). Across the dataset, **25 of 195 countries shift by 10 or more rank places** and 68 shift by 5 or more, depending only on which of the two equally valid measures is used.
- The pattern is systematic by region and income: the European region averages +3.0 pp (older), the Eastern Mediterranean −3.6 pp (younger); high-income countries +1.8 pp, low-income countries −1.6 pp.

7.6.4 Inference and Insights

- Age standardization is doing exactly the job it is supposed to do — and this view makes the size of that job visible. The crude rate is not “wrong”: it is the real service load a country’s health system must actually carry. The age-standardized rate is not “right” either: it describes a population that does not exist. **They answer different questions, and conflating them produces confidently wrong conclusions.**
- The correct use of each is straightforward once separated. To **plan capacity** — how many people need treating in Japan next year — use the crude rate. To **judge health-system performance** — is Qatar’s system doing better than Japan’s? — use the age-standardized rate. Japan’s high crude coverage is substantially a consequence of being old, not of being good.
- Because the high correlation ($r = 0.994$) makes the two measures look interchangeable in aggregate, the risk of using the wrong one is easy to miss — and it is concentrated precisely in the countries that are most demographically unusual, which are often the ones being singled out for comparison in the first place.

7.6.5 Real-World Implications

- **League tables of health-system performance must use age-standardized rates.** A ranking built on crude rates would penalise Qatar by 54 places for the demographic accident of having a young workforce.
- **Budget and workforce planning must use crude rates.** Japan’s clinics face the crude caseload, not the standardized one.
- **For analysts and journalists:** any cross-country chronic-disease comparison that does not state which of the two measures it uses should be treated as unreliable, because the choice can reorder the table.

8 Cross-Cutting Findings

The six views were built as separate tasks, but read together they tell one coherent story about how chronic disease care fails, and the linked dashboard is what makes reading them together possible. Four findings cut across the whole system:

1. **The world treats far more people than it did in 1990, and the gap between rich and poor countries is wider than ever.** Universal improvement and worsening inequality are not contradictory: they are what happens when growth is proportional rather than targeted (Task 3).
2. **Treatment is not care.** Globally, fewer than half of the people treated for hypertension are effectively controlled, and the systems with the worst conversion are not the poorest but the middle-income ones that scaled up prescribing without scaling up follow-up (Tasks 1 and 4).
3. **Men are the under-served sex almost everywhere** — in 97% of countries for hypertension treatment — and the gap is growing. This is very likely a diagnostic gap rather than a treatment gap, and it is invisible to any monitoring framework that reports national averages (Task 2).
4. **How you measure changes who wins.** Income group predicts coverage better than geography does (Task 5), and the choice between crude and age-standardized rates can move a country 54 places in a global ranking (Task 6). Methodological choices in this field are not technicalities; they are policy choices with consequences.

9 Challenges and Limitations

The most significant challenge of the project arrived before any code was written. The WHO Health Inequality Data Repository is a large and inviting dataset, but a completeness audit of all 33 indicators showed that the majority could not support a country-comparative dashboard: financial-hardship and workforce indicators cover only a handful of countries per year, some have no national-level values at all, and others are disaggregated only along dimensions with very thin country coverage. We had to choose between a system with many views that were mostly empty and a system with six views that were completely populated. We chose the latter, and every design decision downstream followed from it. The honest cost of that choice is that this project analyses a *slice* of global health inequality — NCD treatment and control — and cannot speak to the financial-hardship or workforce dimensions that the wider repository gestures at.

The remaining limitations are properties of the data and of our scope, and each bounds what the findings above can be taken to mean:

- **These are modelled estimates, not measurements.** The WHO/NCD-RisC values are statistically modelled from underlying surveys. A country with sparse survey data still receives an estimate for every year, and that estimate carries far more uncertainty than one from a country with regular national surveys. The release we used does not publish uncertainty intervals, so the dashboard cannot draw confidence bands — and small differences between countries (a percentage point or two) should not be over-read.
- **Two indicator families end in different years.** Diabetes runs to 2022, hypertension only to 2019. Any cross-family comparison in the most recent years is therefore comparing different points in time.

We handled this in the interface with an explicit nearest-year fallback (Section 6.3) rather than hiding it, but the underlying asymmetry remains.

- **Sex is binary in the source data.** The repository publishes Male and Female only. Our entire sex-gap analysis inherits that limitation and cannot say anything about non-binary or intersex populations.
- **Income group is a 2025 classification applied retrospectively to 1990–2022.** Every country carries its current World Bank income group for its whole history, but several countries changed group dramatically over the period (the Republic of Korea and Poland being obvious cases). Income-group trend lines in Task 3 therefore mix a genuine performance effect with a composition effect, and the "high-income" line for 1990 describes countries that were not all high-income in 1990. This is a real confound and we flag it rather than correct it, because the repository does not publish historical income classifications.
- **Five small territories are unclassified for income** and fall outside the income analysis entirely.
- **Task 5 cannot drill down to the country level.** Its source table is pre-aggregated into region×income cells, so clicking a bar sets the scope but cannot reveal the individual countries inside it — which, given the large within-cell standard deviations we found, is the most conspicuous functional gap in the dashboard.
- **ΔLeak is computed on crude values only**, so the cascade view inherits the demographic distortion that Task 6 exists to expose. A country with an unusually young or old population has a slightly distorted leak figure.
- **No statistical significance testing.** The dashboard reports differences; it does not test them. Every comparison in this report should be read as descriptive.
- **Performance is adequate rather than scalable.** The full working set is held in memory and re-filtered per callback, with an LRU cache in front of the figure builders. This is comfortable at 72,420 rows and would not be at ten million.

10 Conclusion

This project set out to build a linked visual analytics system for global inequality in diabetes and hypertension care, and to use it to answer a specific question: where are people being treated but not controlled, and how does that gap vary by sex, region, income, and time? The system was delivered as a single-page Dash dashboard with six cross-filtered views over 72,420 verified-complete WHO estimates spanning 195 countries and three decades.

What the dashboard shows is more pointed than "inequality exists." The headline findings are:

- **Care is stratified, not merely uneven.** Diabetes treatment coverage runs from 8.5% in Burkina Faso to 86.0% in Belgium — a 10× ratio — and the ordering of income groups is stable across every indicator we examined.
- **Thirty years of universal progress have widened every absolute gap.** The high-income-to-low-income gap in hypertension effective control grew from 4.4 pp in 1990 to 18.5 pp in 2019, even though low-income countries improved by over 400% in relative terms. Growth rates converged; levels diverged.

- **Treatment is the easy half.** Fewer than half of treated hypertensive patients reach effective control globally, and the worst conversion rates belong to upper-middle-income countries (43.7%) — systems that scaled up prescribing without scaling up follow-up — not to the poorest ones.
- **Men are systematically under-reached.** Women have higher hypertension treatment coverage in 190 of 195 countries, and the gap has widened since 1990 — a signature of a diagnostic rather than a therapeutic failure.
- **Measurement is a policy choice.** The crude-versus-age-standardized decision alone moves Qatar 54 places in a global ranking.

Equally important is what the project demonstrates methodologically. Several of these findings are only visible *because* the views are linked and the metrics are paired. The leakage paradox — that rich countries leak the most in absolute terms while converting the best — is invisible unless leakage and conversion rate are shown together. The demographic confound in Task 6 is invisible unless crude and standardized rates are placed on the same axis. And the decision to scope the project to six verified-complete indicators, made before any code was written, is what allows every claim in this report to be checked against a chart with no gaps in it.

11 Future Work

Several extensions follow directly from the limitations identified above:

- **Country-level drill-down in Task 5.** The most obvious functional gap. Given the large within-cell standard deviations we measured, clicking a region×income bar should reveal the distribution of countries inside it — as a box plot or a dot-strip — rather than only setting the scope. This requires joining the pre-aggregated summary table back to the country-level data.
- **Uncertainty representation.** If a WHO release with credible intervals becomes available, the map and trend views should carry them, so that a 2-point difference between two sparsely surveyed countries is visibly not the same claim as a 2-point difference between two well-surveyed ones.
- **Historical income classification.** Reconstructing each country's World Bank income group *as of each year* would remove the composition confound in the income trend lines and let us separate "poor countries got better" from "countries that got richer moved groups."
- **A diagnosis-coverage indicator.** Our central inference in Task 2 — that the sex gap is a diagnostic gap rather than a treatment gap — is an inference, not a measurement. The WHO repository publishes diagnosis coverage for some countries; adding it would let the dashboard extend the cascade one stage to the left (*diagnosed* → *treated* → *controlled*) and test that inference directly.
- **Persistent, shareable state.** Encoding the six-field filter store into the URL would make any view in the dashboard linkable — important for a tool intended for researchers and policy analysts who need to cite what they saw.
- **A database backend.** If the scope were widened to the full repository, or to sub-national data, the in-memory pandas layer should be replaced with SQLite or DuckDB — the option the proposal held open and that the current data size did not justify.

12 Team Contributions

Table 4: Group 5 — contributions by component.

Component	Member	Roll No.
Data Preprocessing	Stuti Singh	251083
Task 1: Global Coverage Map	Annadevara Yashvanth Chary	240143
Task 2: Sex Gap Analysis	Singampalli Sri Varshith	251110074
Task 3: Temporal Trend Analysis	Mavilla Bharadwaja	251110021
Task 4: Treatment-to-Control Cascade	Talakola Maha Lakshmi Khasha Meganakshtra	241082 240542
Task 5: Regional and Income-Group Comparison	Pyna Hema Lakshmi SatyaSree Sruja BG	240819 240289
Task 6: Age-Standardized vs. Crude Rate Comparison	Unnati Gangurde	231107

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